Adaptive Center-Focused Hybrid Attention Network for Change Detection in Hyperspectral Images

Fenlong Jiang, *Member, IEEE*, Shining Zhang, Mingyang Zhang, *Member, IEEE*, Maoguo Gong, *Fellow, IEEE*, Yu Zhou, Wei Zhao, *Member, IEEE*, and Ziyu Guan, *Member, IEEE*

***Abstract*— Hyperspectral images (HSIs) capture extensive spatial and spectral information, facilitating detailed change detection (CD) of complex land covers. However, the high correlation among spectral data can lead to information redundancy, increasing processing dimensions and introducing irrelevant or detrimental data to CD. To address these challenges, we propose an adaptive center-focused hybrid attention network (ACFHAN) for CD in HSIs. This network adaptively emphasizes the spatial regions and spectral channels most pertinent to CD while suppressing irrelevant information. The architecture establishes an end-to-end mapping from the two HSIs to the change results, featuring multiple center-focused hybrid attention blocks (CFHABs). Each CFHAB integrates two different attention modules, including an adaptive spatial–spectral hybrid self-attention (S2HSA) module that dynamically adjusts spatial–spectral feature weights and a center-focused attention (CFA) module that enhances the area most relevant to the center pixel to be classified. Additionally, to tackle the challenges of expensive labeling, we further designed a multiscale superpixel-based data augmentation method which combines traditional unsupervised and supervised methods to provide sufficient low-cost but high-confidence labeled data for CD. Experimental results across various HSI CD datasets validate the effectiveness of our proposed method.**

***Index Terms*— Center-focused self-attention, change detection (CD), hyperspectral images (HSIs), superpixel segmentation.**

Received 22 October 2024; revised 25 December 2024; accepted 20 January 2025. Date of publication 27 January 2025; date of current version 6 February 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62036006, Grant 62376205, Grant 62133012, Grant 61936006, and Grant 62425605; in part by the Key Research and Development Program of Shaanxi under Program 2024CY2-GJHX-15; in part by the Natural Science Basic Research Program of Shaanxi under Grant 2024JC-YBQN-0633; and in part by the Fundamental Research Funds for the Central Universities, China, under Grant XJSJ24016. *(Corresponding author: Maoguo Gong.)*

Fenlong Jiang, Wei Zhao, and Ziyu Guan are with the Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, School of Computer Science and Technology, Xidian University, Xi'an 710071, China (e-mail: jiangfenlong@outlook.com; ywzhao@mail.xidian.edu.cn; zyguan@xidian.edu.cn).

Shining Zhang, Mingyang Zhang, and Yu Zhou are with the Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, School of Electronic Engineering, Xidian University, Xi'an 710071, China (e-mail: shiningzhang@stu.xidian.edu.cn; omegazhangmy@gmail.com; zhouyu@xidian.edu.cn).

Maoguo Gong is with the Key Laboratory of Collaborative Intelligence Systems, Ministry of Education, School of Electronic Engineering, Xidian University, Xi'an 710071, China, and also with the Academy of Artificial Intelligence, College of Mathematics Science, Inner Mongolia Normal University, Hohhot 010011, China (e-mail: gong@ieee.org).

Digital Object Identifier 10.1109/TGRS.2025.3534881

I. Introduction

With the continuous development of remote sensing technology, an integrated Earth observation system spanning multiple layers, multiple perspectives, and multiple fields from the ground to outer space is taking shape [1]. Increasingly, remote sensing applications are being integrated into modern life, providing essential services in areas such as environmental monitoring [2], disaster assessment [3], and urban planning [4]. Remote sensing change detection (CD) identifies and quantitatively describes changes in surface features by analyzing and comparing multitemporal remote sensing images of the same region. Leveraging the wealth of remote sensing resources for detecting and recognizing these changes helps meet the needs of diverse applications, providing crucial decision-making support for human activities.

Remote sensing images are typically acquired through platforms such as satellites [1], aircraft [5], or unmanned aerial vehicles [5]. Each type of remote sensing image has unique characteristics, requiring specialized analysis for accurate interpretation and processing. Common types include multispectral images [6], [7], synthetic aperture radar (SAR) images [8], [9], and hyperspectral images (HSIs) [10], [11], [12]. Multispectral images usually contain 3 to 10 spectral bands, while SAR images use microwave frequencies for high-resolution imaging. In contrast, HSIs have an "image cube" structure that integrates spatial and spectral dimensions, facilitating more detailed material analysis and target detection [13]. These features render HSIs particularly valuable for CD.

The rich spectral information in HSIs is crucial for distinguishing between different types of ground objects. However, the high spectral correlation among adjacent bands leads to significant spectral redundancy. Early methods, like principal component analysis (PCA) [14], were used for dimensionality reduction, consolidating most of the relevant change information into a few principal components, effectively reducing spectral redundancy. However, this approach does not directly capture the specific nature of changes, and the dimensionality reduction process can result in the loss of critical change-related information, thereby affecting the accuracy of CD. Recent advances in deep learning have led to numerous CD methods using convolutional neural networks (CNNs) [15], [16] to extract both spatial and spectral features that are representative and discriminative. The strong feature representation

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

capabilities of deep architectures enable automatic extraction and representation of relevant features. However, CNNs are inherently better suited for capturing local spatial features and struggle with capturing long-range dependencies and global information. To overcome these challenges, more recent CD methods [17], [18] have integrated attention methods within neural networks to enhance convolutional features. These attention modules dynamically adjust the network's focus by assigning importance weights to different spatial regions, allowing for more adaptive feature selection. However, despite these advances, attention methods still depend on convolution-based feature extraction, which limits their ability to handle long-range dependencies effectively. Many challenges remain to be addressed.

1) HSIs often contain numerous redundant bands, and convolutional operations lack the ability to adaptively differentiate between significant and redundant bands. Consequently, convolutional kernels may treat all bands uniformly when processing redundant information, leading to reduced feature extraction efficiency and degraded model performance.
2) Ground-truth labels for target objects in HSIs CD are typically provided at the pixel level. Many patch-based methods assume that all pixels within a neighborhood share the same characteristics as the central pixel for computational simplicity. However, this assumption fails at boundaries [19], complicating the accurate extraction of features at these critical regions.
3) HSIs CD requires pixel-level ground truth, yet a single HSI can contain tens to hundreds of thousands of pixels, making manual labeling a time-consuming and costly process, ultimately limiting the availability of labeled samples [20], [21].

To address these challenges, this article proposes an adaptive center-focused hybrid attention network (ACFHAN), designed to enhance strongly correlated features correlated with CD while suppressing irrelevant ones. ACFHAN constructs an end-to-end mapping from HSI image patch pairs to patch center pixel CD results. Specifically, ACFHAN is primarily composed of multiple center-focused hybrid attention blocks (CFHABs), which integrate two attention modules: an adaptive spatial–spectral hybrid self-attention (S2HSA) module and a center-focused attention (CFA) module. Given the redundancy and similarity of spectral information across many bands of HSIs, such similar data often fails to provide additional insights, diminishing overall data utility. To address this, the S2HSA module is proposed to assign weight coefficients to each pixel by computing spatial and channel self-attention (CSA) scores, effectively highlighting the most relevant spatial and spectral information. To maintain the integrity of the original data, these attention scores are added back to the feature map before it is passed to subsequent layers. The CFA module, designed to tackle boundary ambiguity, measures the self-correlation within a patch by calculating cosine similarity-based attention scores between the central pixel and its neighboring pixels. This enables more accurate extraction of boundary features and has the potential to desensitize the size of the patch.

In addition, to reduce the reliance on manual labeling and decrease the cost of acquiring labeled samples for HSI data, we introduce a data augmentation method based on superpixels. This method combines superpixel segmentation [22] with traditional unsupervised and supervised algorithms to enhance the training dataset by generating a substantial number of low-cost, high-confidence labeled samples. The main contributions of this article are summarized as follows.

1) We propose a novel ACFHAN network for HSIs CD that adaptively enhances CD-related features through the integration of multiple CFHAB modules composed of S2HSA and CFA modules while effectively suppressing irrelevant information.
2) To better obtain boundary information, we propose a cosine similarity-based CFA module that can adaptively focus on the pixels most relevant to the central pixel while also reducing sensitivity to patch size to some extent.
3) We introduce a training strategy based on superpixels, combining superpixel segmentation with traditional unsupervised and supervised algorithms to augment labeled data effectively.

The remainder of this article is organized as follows. Section II reviews related work, Section III presents the proposed methodology, Section IV discusses experimental results on multiple datasets, and Section V concludes the article with a summary of our findings.

II. Related Work

A. HSI-Based CD

A variety of algorithms have been developed for HSIs CD, evolving from basic algebraic approaches to sophisticated feature extraction techniques, and from utilizing purely spectral to integrating spatial–spectral information. Traditional methods include algebra-based approaches, transformation-based methods, spectral analysis methods, and postclassification methods. Algebra-based methods [23] primarily measure the magnitude and direction of change by calculating the distance between multitemporal spectral vectors. Considering the spectral variability of HSIs, Liu et al. [24] iteratively analyze change vectors to achieve a more accurate assessment of change status. Transformation-based methods, on the other hand, involve converting HSIs into a lower dimensional feature space to distinguish between changed and unchanged areas. PCA [25] is one of the most popular subspace learning algorithms. Wu et al. [26] introduced a spectral analysis-based CD method for HSIs, which detects changes by examining spectral distances or differences in shape between all bands of pixel pairs from two temporal images. Postclassification methods [27] typically use two or more datasets to create Change Maps (CMs). However, these methods can overlook the correlation between bitemporal images when classification accuracy is insufficient, leading to suboptimal detection results.

With the advent of CNN [28], deep learning-based methods have gained significant traction in HSIs CD [29]. The strength of these models lies in their ability to deeply integrate both spatial and spectral features, leading to notable improvements

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

in performance. These methods can be broadly categorized into unsupervised and supervised approaches.

Unsupervised CD methods, on the other hand, do not rely on labeled data. Instead, they typically map data to a new feature space through specific constraint functions or network loss functions, maintaining relevance in scenarios where labeled data is scarce. For example, Du et al. [30] proposed an unsupervised Deep Slow Feature Analysis network, which leverages the powerful feature extraction and projection capabilities of deep networks. However, it processes the 2-D imaging space and spectral response dimensions in a mixed manner, which is insufficient to extract discriminative features and leads to limited accuracy. Therefore, Yang et al. [31] proposed a framework using randomly initialized models with spatial–channel augmentation, which explores the spatial correlations and discriminative spectral features of multitemporal HSIs separately. Despite the success of unsupervised algorithms, they perform poorly in some complex scenarios, hindering their widespread adoption.

Most supervised algorithms are classification-based, where the model is trained to output a binary result, indicating whether a pixel has changed. For instance, Zhan et al. [32] input HSIs into a Siamese network CNN architecture to extract pixel-level shallow spatial–spectral feature vectors, and then compute the Euclidean distance between the two vectors to represent their similarity. However, this method focuses on adjusting the network architecture or adding extra handcrafted modules to improve the CNN, resulting in a more complex network structure and significantly increased computational costs. To address this issue, Wang et al. [33] proposed an end-to-end residual self-calibration network to improve the accuracy of HSIs CD. This method adaptively constructs spatial and spectral interdependencies for each spatial location, allowing for better utilization of spatial information while reducing additional parameters and lowering computational complexity. However, the high cost of data labeling has always been a major issue faced in HSIs CD. Therefore, considering the trade-offs for the HSIs CD task, semisupervised methods may be more suitable.

B. Attention-Based HSIs CD Methods

Attention-based methods refer to the process by which the human nervous system filters out irrelevant or nonessential information, selectively focusing on specific stimuli in the presence of complex sensory data. In recent years, attention-based methods have gained significant traction in the field of CD, particularly in HSIs CD. These methods allow models to concentrate on the most relevant data features, enhancing both feature extraction and change recognition.

In HSIs CD algorithms, attention-based methods were initially applied to the extraction of spectral features. Haut et al. [34] integrated attention-based methods into ResNet to better capture the spectral information within the data. Mou and Zhu [35] employed a gating mechanism to adaptively recalibrate spectral bands by selectively emphasizing informative bands and suppressing less useful ones. Fang et al. [36] proposed an end-to-end 3-D dense convolutional network that introduced a spectral attention-based method to enhance the discriminability of spectral features, thereby improving the model's classification performance.

Although the aforementioned methods have shown promising results, relying solely on spectral information for CD may overlook spatial neighborhood variations or noise, leading to false detections or missed changes. A method that incorporates both spatial and spectral attention would better capture fine-grained changes in HSIs while also enhancing the model's generalization ability and robustness. Wang et al. [37] use attention-based method to emphasize channels and spatial positions with abundant information while suppressing those with less, thereby adaptively enhancing spectral–spatial features. But it struggles with detecting subtle changes in bitemporal images, as the decision boundary tends to be distorted by strongly varying regions, thus overshadowing smaller changes. To tackle this, Li et al. [38] proposed a Superpixel-by-Superpixel Clustering Framework, which reduces confusion between different types of changes, ultimately minimizing their impact on the decision boundary. The same way, Dong et al. [39] capture subpixel-level correlation differences in a hierarchical manner, enabling the detection of subtle changes. Recognizing the limitations of multiscale features that combine both changed and unchanged components, Luo et al. [40] proposed a Multiscale Diff-Changed Feature Fusion Network, which improves the representation of fine change components by learning them across different scales in bitemporal HSIs.

To further improve the model's ability to capture subtle changes, a more refined attention method should focus more precisely on the central area of the image. In this context, Gong et al. [41] proposed a Gaussian-based attention method, which allows patch-based models to concentrate more effectively on the central region of a patch. By employing a Gaussian distribution, this approach increases the weight assigned to the center area, enhancing the model's attention to central pixels in CD tasks. However, while the Gaussian approach successfully emphasizes the center, it lacks the flexibility to capture nonlocal information that, although distant from the center, may still be relevant to the central pixel. Additionally, the Gaussian function can inadvertently include nearby pixels that are irrelevant to the detected changes, introducing noise and reducing detection accuracy. Therefore, the challenge remains to balance attention between the center and important nonlocal information, an issue that current methods have yet to fully resolve.

III. Proposed Method

The proposed method for HSIs CD consists of two key components: a feature extraction network and a superpixel-based label augmentation strategy. The feature extraction network is designed to extract CD-related features from HSIs, improving the representation of relevant features and minimizing interference from irrelevant information, while the label augmentation component aims to expand high-confidence labeled samples in a low-cost manner. Below, we provide a detailed explanation of each part of the proposed model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

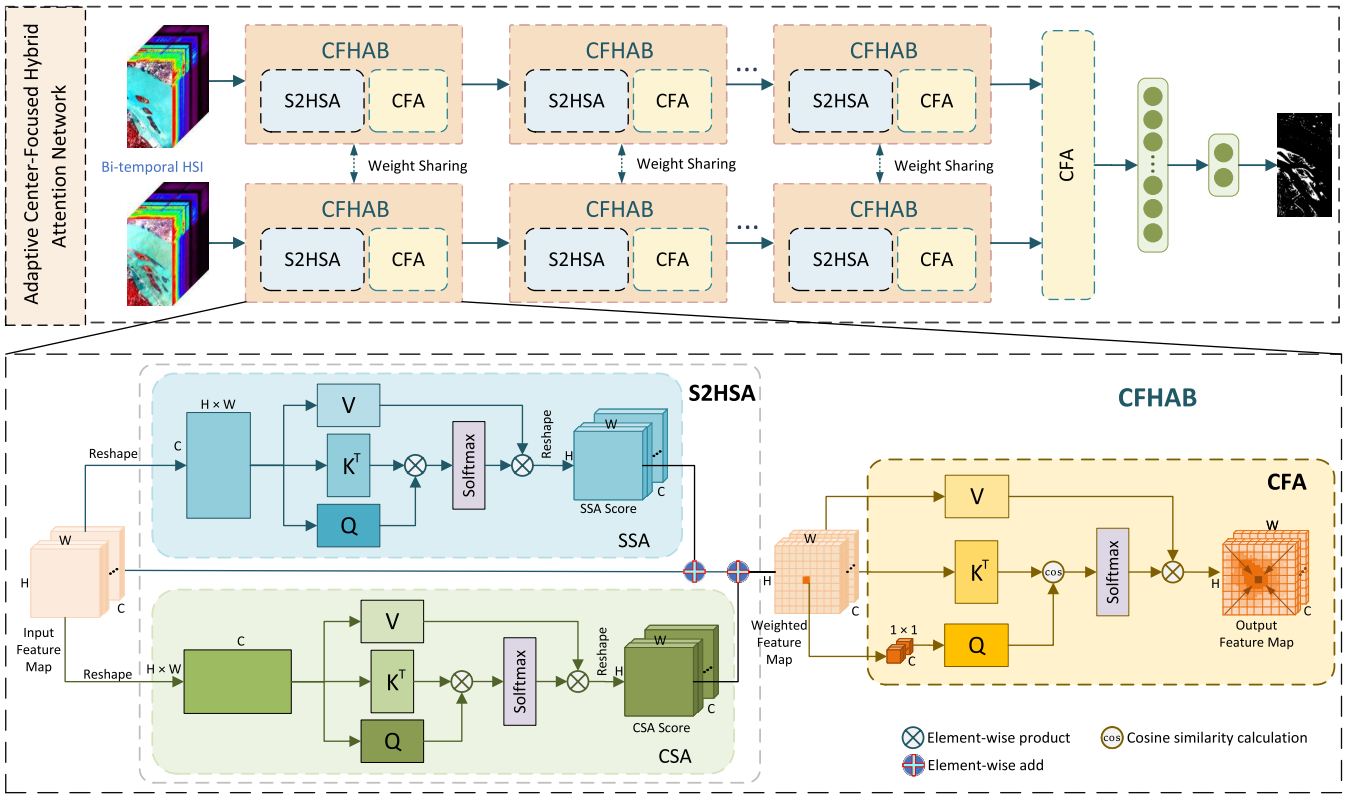


Fig. 1. Architecture of the proposed ACFHAN. The ACFHAN employs several Siamese CFHABs to continuously capture the self-correlation of feature maps. The CFHABs primarily consist of the S2HSA for extracting spatial and channel features, and the CFA for adaptively computing the potential impact of each surrounding pixel on the central pixel in a point-to-point manner, enhancing the focus on the central pixel.

A. ACFHAN

For CD tasks, many deep learning-based methods have already achieved significant success [29], [42]. However, most of these methods use an approach where the difference map is computed first [40], followed by feature extraction. This approach focuses directly on change information, optimizing feature extraction for change areas, but it may overlook some details in the original images since the difference map only retains change-related information. Additionally, directly using the difference map can be sensitive to noise; poor image quality can significantly impact the quality of the difference map and, consequently, the feature extraction process. To address these issues, we adopted a dual-branch Siamese network structure. In this approach, features from the HSIs are extracted first before computing the differences. This method preserves more original information and details, leading to more comprehensive feature extraction. It also provides more robust feature representations, reducing the impact of noise and improving CD accuracy.

As illustrated in Fig. 1, the ACFHAN employs several Siamese CFHABs to continuously capture the self-correlation of feature maps. The CFHABs primarily consist of the S2HSA for extracting spatial and channel features, and the CFA for minimizing interference from irrelevant information around the patches while enhancing the representation of change information. We will now provide a detailed description of these modules.

1) S2HSA: Traditional convolutional operations have limitations in capturing long-range dependencies and global information due to the size and stride of the convolutional kernels. In contrast, the self-attention effectively captures long-range dependencies and global information, allowing the model to establish more flexible connections between different spatial regions without being constrained by a fixed receptive field [43]. Therefore, we adopt the S2HSA module, which incorporates self-attention-based spatial self-attention (SSA) and CSA, to directly capture global spatial and channel dependencies in HSIs, effectively considering the relationships between distant pixels. This method dynamically adjusts the regions of focus to better retain crucial image information and prevent feature loss. Specifically, based on the generated pseudo-labels, one input patch $P_1 \in \mathbb{R}^{C\times H\times W}$ is selected from the first HSI, and another input patch $P_2 \in \mathbb{R}^{C\times H\times W}$ is selected from the second HSI. Here, H, W, and C represent the height, width, and number of channels of the patches, respectively. The operations on patches within the network are illustrated in Fig. 1. First, we need to rearrange P_i, adjusting the tensor to a format suitable for SSA, denoted as S_i, and a format suitable for CSA, denoted as T_i. The process is as follows:

$$S_i = \text{reshape}(P_i, (C, H \times W)) \tag{1}$$

$$T_i = \text{reshape}(P_i, (H \times W, C)) \tag{2}$$

where i represents the index of the HSIs, $\text{reshape}(P_i, (C, H \times W))$ and $\text{reshape}(P_i, (H \times W, C))$ denote rearranging the input tensors P_i into tensors S_i, T_i with the shape $(C, H \times W)$ and $(H \times W, C)$. Next, we can perform linear transformations on S_i, T_i separately to obtain their corresponding matrices query Q_i, key K_i, and value V_i required for the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

self-attention. Finally, the associations between the input features are computed using the self-attention and the feature maps are described by the following calculation:

$$\text{Atten}_i(Q, K, V) = \text{softmax}\left(\frac{Q_i {K_i}^T}{\sqrt{d_k}}\right) V_i \tag{3}$$

where softmax represents the normalization process. The computation method for Q_i, K_i, and V_i is as follows:

$$\begin{cases} Q_i = \text{linear}(S_i; W_i^Q) \\ K_i = \text{linear}(S_i; W_i^K) \\ V_i = \text{linear}(S_i; W_i^V) \end{cases} \tag{4}$$

where linear denotes the linear transformation. S_i represents that we compute Q_i, K_i, and V_i corresponding to S_i. We can also replace S_i with T_i, indicating that we will compute Q_i, K_i, and V_i corresponding to T_i. W_i^Q, W_i^K, and W_i^V are the corresponding weight matrices.

After computing the corresponding Atten_{S_i} and Atten_{T_i} for CSA and SSA, the matrixes are restored to match the original size of the feature map and applied to the original feature maps $F_{i,\text{ori}}$ as follows:

$$F_i = F_{i,\text{ori}} + \text{Atten}_{T_i} + \text{Atten}_{S_i} \tag{5}$$

where F_i represents the weighted feature map. Building on this, the multihead self-attention method can simultaneously learn information from various subspaces, allowing it to capture a wide range of features and relationships within the input sequence. This capability enhances the model's robustness to diverse inputs, enabling us to further improve the model by incorporating multihead self-attention.

2) CFA: In patch-based CD algorithms, the classification of a target pixel relies not only on its own features but also significantly on the information from neighboring pixels. However, the relationships between the surrounding pixels and the central pixel are not always consistent. Therefore during feature extraction, the relationship between the central pixels $CP_1 \in {P_1}^{C\times 1\times 1}$ and $CP_2 \in {P_2}^{C\times 1\times 1}$ and their surrounding pixels are typically evaluated to emphasize the central pixel's features and ensure its predominant role in classification. To address this, we propose an attention module based on cosine similarity, which computes the potential impact of each surrounding pixel on the central pixel in a point-to-point manner. The specific approach is outlined as follows:

$$Fa_i(Q, K, V) = \text{softmax}\left(\frac{Q_i \cdot {K_i}^T}{\|Q_i\| \left\|{K_i}^T\right\|}\right) V_i \tag{6}$$

where the calculation methods for Q_i, K_i, and V_i are as follows:

$$\begin{cases} Q_i = f_i^Q(CP_i; W_i^Q) \\ K_i = f_i^K(P_i; W_i^K) \\ V_i = f_i^K(P_i; W_i^V) \end{cases} \tag{7}$$

where $f_i^Q(\cdot)$, $f_i^K(\cdot)$, and $f_i^V(\cdot)$ represent 1×1 convolution operations. Subsequently, the attention scores for the central pixel at each position are expanded across all spectral channels to create the final feature extraction map. The specific method is as follows:

$$F_i = F_i \cdot Fa_i(Q, K, V). \tag{8}$$

By sequentially stacking multiple attention modules, features are extracted and the central point's features are emphasized, creating a feature map for each remote sensing image. These feature maps from different time periods are then subtracted to produce a difference map. However, this subtraction process may diminish the focus on the central pixels. To address this, a CFA module is applied again to reinforce the central point's weight in the difference map before classification. Given that direct linear classification can be computationally demanding, we first perform convolutional operations to reduce the feature dimensions, thereby making the classification of the feature map more efficient and manageable.

B. Superpixel-Based Semisupervised Sample Augmentation Strategy

As shown in Fig. 2, to address the challenge of limited labeled samples during network training, we propose a superpixel-based semisupervised sample augmentation strategy. First, superpixel segmentation maps are generated for each HSI and fused to produce a unified superpixel map with consistent boundaries. This map is then combined with CD results obtained from both unsupervised and supervised classification algorithms. High-confidence outcomes are selected to augment the label set. In the following, we elaborate on this strategy from two key aspects: superpixel segmentation and pseudo-label generation.

1) Superpixel Segmentation: In CD, regions of change are often represented by image blocks rather than individual pixels, making superpixel segmentation a useful technique. Superpixel segmentation [44] groups pixels with similar characteristics into larger, homogeneous "superpixels," retaining key information while preserving object boundaries for further processing tasks. We apply the simple and efficient SLIC algorithm [45] to segment the HSIs, as it effectively produces compact superpixels with well-preserved boundaries.

While SLIC performs well for simple images, it struggles with complex images affected by noise, brightness variations, and low contrast, which can lead to suboptimal segmentation [45]. Additionally, the superpixels generated from HSIs during segmentation are often misaligned, complicating the extraction of corresponding information for comparison in later feature extraction stages. To address this, we adopt a superpixel boundary fusion strategy. As shown in Fig. 2, SLIC is applied to both HSIs, generating two superpixel maps, X_1 and X_2, composed of multiple superpixel blocks

$$X_i = \bigcup_{n=1}^{N} R_{i,n}, i = 1, 2 \tag{9}$$

where N denotes the total number of superpixel blocks, $R_{i,n}$ represents each superpixel block with highly similar pixels, and n is the label distinguishing different superpixels.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

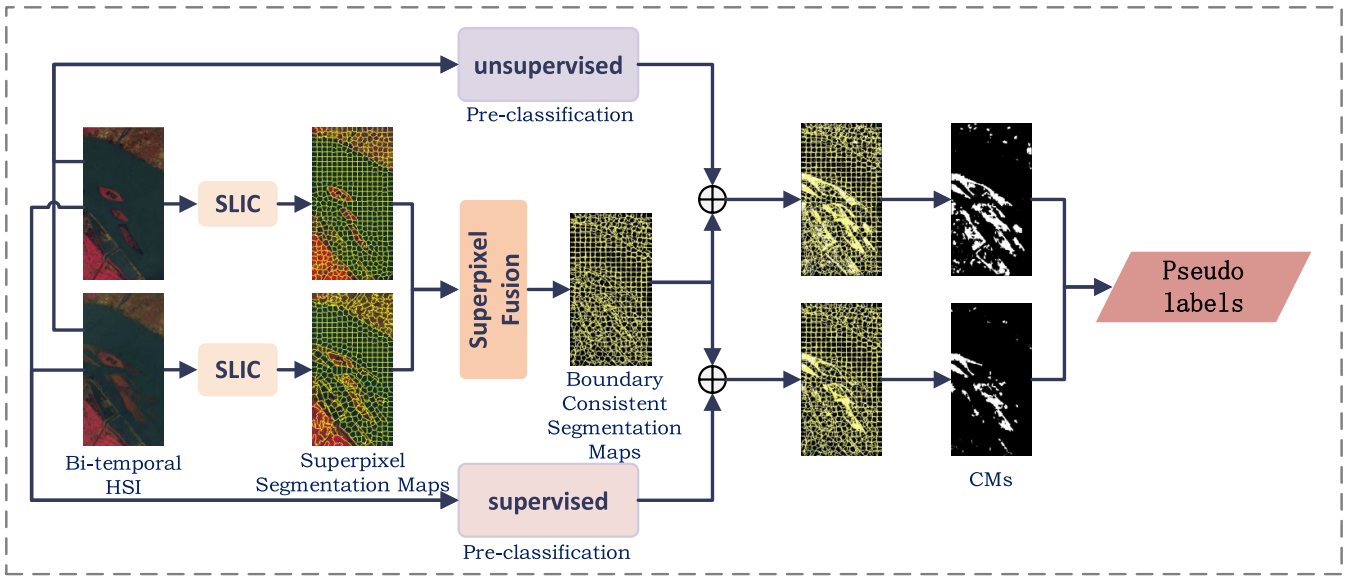


Fig. 2. Illustration of the superpixel-based semisupervised sample augmentation strategy.

To fuse these maps, we determine the relative positions of corresponding superpixel regions. If superpixel blocks R_1 and R_2 overlap, we assign a common label L to the overlapping region, while nonoverlapping parts receive distinct labels, generating a new superpixel map. The label determination is as follows:

$$X_{\text{HSIs}} = \begin{cases} L, & R_1(\Omega) \cap R_2(\Omega) \neq \varnothing \\ L', & \text{else} \end{cases} \tag{10}$$

where X_{HSIs} represents the newly segmented superpixel map, Ω is the region occupied by the superpixel, and L and L' are the labels for overlapping and nonoverlapping regions. This operation results in superpixel maps R_n with consistent boundaries, laying the foundation for pseudo-label generation.

2) Pseudo-Label Generation: After generating a superpixel segmentation map with consistent boundaries, this map is matched with CD results from both unsupervised and supervised classification algorithms to produce unsupervised and supervised CMs based on superpixels. Subsequently, the results from both methods are combined, and high-confidence outcomes are selected as pseudo-labels to augment the label set. We elaborate on this strategy in detail next.

Hyperspectral remote sensing is a technology that continuously captures images of ground objects across very narrow and continuous spectral channels. In the visible to shortwave infrared bands, its spectral resolution can reach the nanometer scale, with the number of spectral channels typically numbering in the dozens or even hundreds. Spectral features are crucial for HSI analysis, especially for CD. We employ the CVA algorithm [23] to analyze the spectral features of HSIs. This algorithm identifies changes in ground objects by comparing spectral features from two different time points and utilizing change vectors. The formula is as follows:

$$F_{\text{HSIs}} = \sqrt{\left(\sum_{\text{b}=1}^{B}\left(\overset{b}{I}_1(x) - \overset{b}{I}_2(x)\right)\right)^2} \tag{11}$$

where x denotes the pixels in each HSI, B represents the number of spectral bands, and b indicates the index of the spectral band in the HSIs. $I_1^b(x)$ and $I_2^b(x)$ are the pixel values from the original HSIs. F_{HSIs} refers to the spectral difference features extracted from HSIs.

Based on these spectral differences, we classify pixels as changed or unchanged using the unsupervised Otsu thresholding algorithm [46], resulting in a preliminary CM CM_{p_1}. However, since this method above is unsupervised, it may lack specificity and accuracy for certain tasks. To enhance accuracy, we incorporate supervised information through the KNN algorithm [47], resulting in the second CM CM_{p_2}. We then match CM_{p_1} and CM_{p_2} with the superpixel map X_{HSIs}, calculate the percentage of changed pixels within each superpixel for each CM, and apply a threshold τ to determine the overall change status of each superpixel block. When the proportion of change pixels exceeds this threshold, we can conclude that the superpixel block has experienced a change; conversely, if the proportion falls below the threshold, we do not think this superpixel block has changed. This process generates an unsupervised superpixel-based CM CM_1 and a supervised superpixel-based CM CM_2, outlined as follows:

$$E_{d,n}^{\text{change}} = \frac{\sum_{x \in R_n} CM_{p_d}(x)}{M}, d = 1, 2 \tag{12}$$

where d represents the index of the CMs, $E_{d,n}^{\text{change}}$ denotes the percentage of changed pixels within each superpixel for each CM, and M indicates the total number of pixels within each superpixel block. The threshold selection method is as follows:

$$\underset{d, x \in R_n}{CM}(x) = \begin{cases} 1, & E_{d,n}^{\text{change}} \geq \tau \\ 0, & \text{else} \end{cases} \tag{13}$$

where $CM_{d, x \in R_n}$ represents the CM block of CM_1, CM_2, which can be expressed as follows:

$$\underset{d}{CM} = \bigcup_{n=1}^{N} \underset{d, x \in R_n}{CM}, d = 1, 2. \tag{14}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

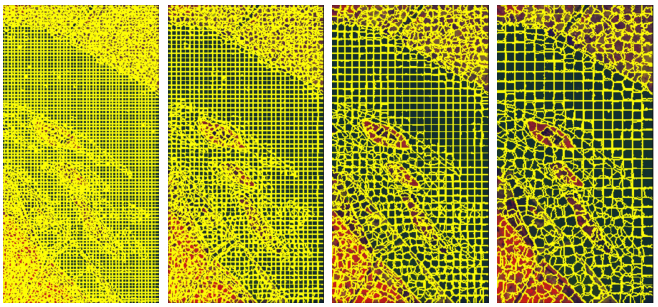


Fig. 3. Part of the segmentation result maps. The number of superpixel regions is (a) 5000, (b) 2500, (c) 1000, and (d) 500.

Finally, we integrate the unsupervised and supervised results to calculate the probability of a pixel being classified as change or unchange at corresponding positions in two results, emphasizing that only the sample points classified as change and unchanged by both methods possess a high level of confidence. The final set of selected labels $\text{Label}_{\text{selected}}$ is expressed as follows:

$$\text{Label}_{\text{selected}} = CM_1 \cap CM_2 \tag{15}$$

where $\cap$ represents the intersection operation.

However, although the aforementioned operations can identify some high-confidence samples, the fixed scale of superpixels and imprecise boundary segmentation still introduce significant uncertainties. Therefore, a multiscale fusion approach is worth considering, as it combines results from different scales, which can further enhance the quality in the selected sample points. The significant variation in the sizes of ground objects in HSIs, as shown in Fig. 3, reveals that increasing the number of superpixels leads to smaller superpixel blocks, thereby reducing the amount of spatial structure information captured within each region. Conversely, reducing the number of superpixels results in larger superpixel blocks, which compromises the boundary details of smaller objects and leads to less precise boundaries. This presents a challenge: fewer superpixels result in the loss of fine image details, while a higher number of superpixels may increase sensitivity to noise and texture, ultimately affecting segmentation quality. To ensure consistency and reliability in the superpixel-based label augmentation process, we utilize multiple superpixel counts to segment the HSIs, generating superpixel maps of varying granularities. Then we can obtain CMs based on superpixels of different scales, and subsequently integrate these CMs as follows to derive more robust and accurate pseudo-labels:

$$\text{Label}_{\text{selected}} = \text{Label}_1 \cap \text{Label}_2 \cap \cdots \cap \text{Label}_d \tag{16}$$

where Label_d represents the label selected on the dth scale based on (15).

Following this process, we obtain a reliable set of samples from the HSIs, including both changed and unchanged regions. These samples are then used to train the proposed ACFHAN network. During training, the network learns to effectively distinguish between changed and unchanged pixels, ultimately producing the final CD results.

C. Learning Process of the ACFHAN

The process begins by feeding the high-confidence training samples constructed in Section III-B into the ACFHAN network for forward propagation, which generates predicted results. Next, we employ the cross-entropy loss function to assess the discrepancy between the predicted results and the true labels [39], [40], [41], [42]. The model parameters are then updated through the backpropagation algorithm to minimize the loss value. This iterative process allows the model to continuously refine its learning of effective feature representations and decision boundaries until the loss converges, ultimately achieving optimal performance on the validation set. The formula for the cross-entropy loss function is as follows:

$$\text{Loss} = -\frac{1}{m}\sum_{i=1}^{m}\left(y_i \log y_p + (1-y_i)\log(1-y_p)\right) \tag{17}$$

where m represents the number of samples, y_i denotes the true label of the sample, and y_p represents the predicted classification result.

IV. Experiments

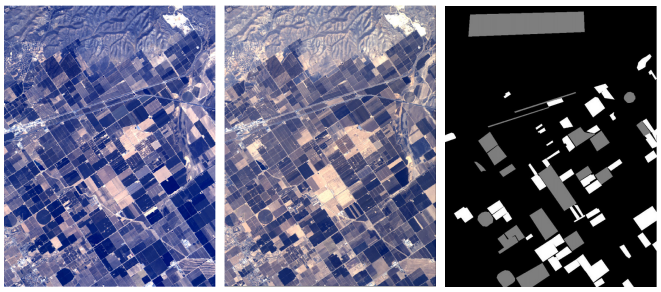
To validate the effectiveness of the proposed method, we conducted experimental analyses on three public datasets. We compared our approach with several CD algorithms and demonstrated the superiority of our method. This section introduces the datasets and experimental settings used, presents the experimental results, and includes an analysis of the algorithm's robustness to different patch sizes. We also performed rigorous ablation studies to assess the effectiveness of the various modules and methods proposed in our algorithm.

A. Datasets

To assess the performance of our method in real-world scenarios, we conducted experiments on four publicly available hyperspectral datasets: the *SantaBarbara Dataset*, the *BayArea Dataset* [48], the *River Dataset* [49], and the *China Dataset* [50]. Each dataset comprises one reference map and two HSIs captured over the same area at different times. As shown in Figs. 4(c) and 5(c), the reference maps for the first two datasets are tricolor, where white represents areas of change, gray represents areas of no change, and black indicates unknown areas. In contrast, as shown in Figs. 6(c) and 7(c), the reference map for the River dataset and China dataset is binary, where black represents unchanged pixels and white represents changed pixels.

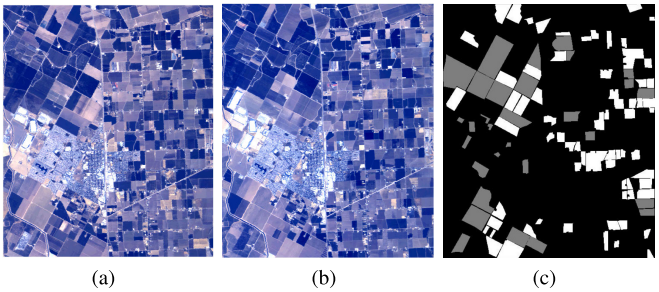
1) SantaBarbara Dataset: As depicted in Fig. 4, the two HSIs in the SantaBarbara dataset were captured by the AVIRIS sensor in 2013 and 2014, covering the SantaBarbara region of California. AVIRIS, developed by NASA, is an imaging spectrometer capable of capturing detailed spectral information for each pixel, allowing for the detection of the physical and chemical properties of surface materials. Consequently, analyzing these two images enables the identification of urban surface changes over a one-year period, which is critical for urban planning and management. Each image in the SantaBarbara dataset has a spatial resolution of 984 $\times$ 740 pixels and contains 224 spectral bands.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.



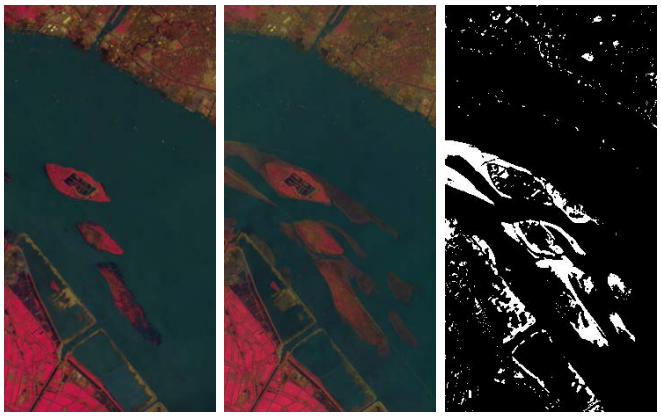
(a) (b) (c)

Fig. 4. SantaBarbara dataset. (a) Image acquired in 2013. (b) Image acquired in 2014. (c) Reference map.



(a) (b) (c)

Fig. 5. BayArea dataset. (a) Image acquired in 2013. (b) Image acquired in 2015. (c) Reference map.

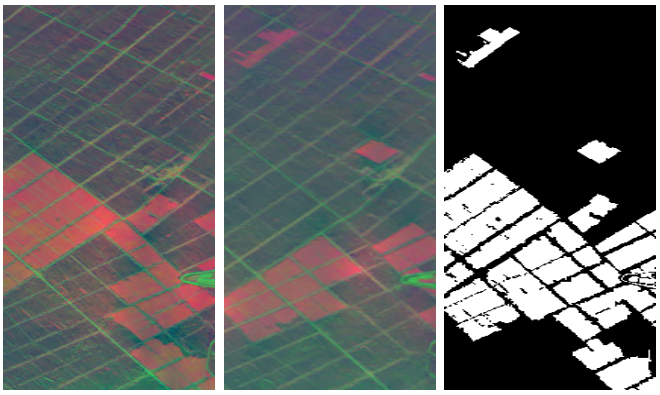


(a) (b) (c)

Fig. 6. River dataset. (a) Image acquired on May 3, 2013. (b) Image acquired on December 31, 2013. (c) Reference map.

2) BayArea Dataset: Similarly, as shown in Fig. 5, the BayArea dataset reflects changes in the Patterson urban area and surrounding farmland between 2013 and 2015. This dataset was also captured by the AVIRIS sensor, with each image having a resolution of 600 × 500 pixels and containing 224 spectral bands. The spectral range spans from 400 to 2500 nm, covering both the visible and shortwave infrared regions. The corresponding reference map is shown in Fig. 5(c).

3) River Dataset: The River dataset covers river areas in Jiangsu Province, China, as shown in Fig. 6. This dataset was acquired by the Earth Observing-1 (EO-1) Hyperion sensor on May 3, 2013, and December 31, 2013. Each image has a resolution of 463 × 241 pixels. The spectral range spans from 0.4 to 2.5 μm with a spectral resolution of 10 nm and a spatial resolution of 30 m, comprising 242 spectral bands in total. After removing low-quality bands, 198 bands remain available for use. This dataset primarily highlights changes in the river channels.



(a) (b) (c)

Fig. 7. China dataset. (a) Image acquired on May 3, 2006. (b) Image acquired on April 23, 2007. (c) Reference map.

4) China Dataset: The China dataset is derived from a farmland located near Yuncheng, Jiangsu Province, China, and consists primarily of soil, river, trees, buildings, roads, and agricultural fields. It was acquired by the Hyperion sensor on May 3, 2006, and April 23, 2007, respectively. The dataset captures changes related to land cover types and rivers, as shown in Fig. 7. The two HSIs in the dataset each have a size of 420 × 140 pixels, with a spatial resolution of 30 meters. The dataset includes 242 spectral bands, spanning the spectral range of 0.4–2.5 μm. After removing noisy and water absorption bands, 154 spectral bands are retained for CD analysis.

B. Evaluation Criteria

This article employs four widely used evaluation metrics to assess the CD performance of the proposed algorithm: overall accuracy (OA), Kappa coefficient (KC), overall error (OE), and $F1$ score. To compute these metrics, we first need to construct the confusion matrix to determine the values of true positive (TP), true negative (TN), false positive (FP), and false negative (FN). These values represent the number of pixels correctly classified as changed areas, correctly classified as unchanged areas, incorrectly classified as changed areas, and incorrectly classified as unchanged areas, respectively. With these values, the following evaluation metrics can be derived.

1) Overall Accuracy: This metric evaluates the overall performance of the classification model, defined as the proportion of correctly classified samples to the total number of samples

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{18}$$

2) Kappa Coefficient: This metric assesses the agreement between the model's classification results and random classification results, particularly in cases of imbalanced classes. A higher KC [51] indicates better classification performance. The formula for calculating KC is as follows:

$$\mathrm{KC} = \frac{\mathrm{OA} - \mathrm{PRE}}{1 - \mathrm{PRE}} \tag{19}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTITATIVE EVALUATION OF EXPERIMENTAL RESULTS OBTAINED BY DIFFERENT METHODS ON FOUR DATASETS

Methods	Supervised	DL	SantaBarbara				BayArea				River				China			
			OA	OE	KC	F1	OA	OE	KC	F1	OA	OE	KC	F1	OA	OE	KC	F1
CVA	-	-	87.12	17078	0.7320	83.96	87.61	9000	0.7534	87.45	92.16	8744	0.6272	66.81	87.49	7354	0.6964	78.75
RCVA	-	-	86.74	17576	0.8226	83.22	87.90	8789	0.7598	87.46	94.65	5972	0.6760	70.54	87.67	7250	0.6907	77.25
C2VA	-	-	83.20	22265	0.6503	79.03	84.99	10901	0.7022	84.39	92.76	8082	0.6601	69.78	87.51	7342	0.7009	78.92
TDRD	-	-	86.06	18483	0.7194	84.13	88.34	8473	0.7671	88.51	92.18	8726	0.6359	67.64	88.74	6619	0.7321	81.22
PACKeans	-	-	77.16	30277	0.5320	72.72	84.76	11069	0.6971	84.36	95.56	4955	0.7124	73.66	87.29	7473	0.6901	77.80
SVM	✓	-	93.21	8999	0.8568	91.20	92.58	5388	0.8516	92.80	92.42	8463	0.6504	68.96	89.69	6061	0.7651	84.13
KPCAMNet	-	✓	87.93	16000	0.7495	85.07	89.90	7336	0.7990	89.77	94.42	6228	0.6941	71.46	88.10	7000	0.7118	79.49
DCVA	-	✓	79.21	27564	0.5313	66.96	82.48	12726	0.6546	80.62	88.47	12866	0.2466	30.94	82.70	10175	0.5460	65.07
SCIU	-	✓	89.09	14461	0.7718	86.20	87.87	8808	0.7590	87.54	89.34	11900	0.4874	54.34	88.74	19857	0.7280	80.68
GETNET	✓	✓	89.67	13688	0.7800	86.27	84.42	11317	0.6862	85.67	94.72	5887	0.7073	73.61	89.46	6199	0.7643	84.32
TDSSC	✓	✓	93.29	8891	0.8570	91.04	94.37	4093	0.8876	94.39	95.93	4537	0.7598	78.21	85.10	8761	0.6828	79.64
DSFANet	✓	✓	86.86	17549	0.7174	82.24	63.37	26604	0.2800	58.34	94.61	6018	0.6645	69.41	61.55	22607	0.5812	35.49
S^2AN	✓	✓	93.46	8664	0.8621	91.52	95.05	3594	0.9012	95.14	96.67	3771	0.8010	81.92	90.00	5879	0.7555	82.45
MSCSCNet	✓	✓	95.28	6252	0.9008	93.93	94.99	3292	0.8983	95.53	93.84	6873	0.5893	62.27	89.94	5913	0.7618	83.37
DCENet	✓	✓	94.21	7675	0.8781	92.54	95.20	3153	0.9011	95.90	95.28	7314	0.6587	68.35	92.35	4498	0.8228	87.86
Ours	✓	✓	**96.38**	**4801**	**0.9245**	**95.46**	**96.40**	**2364**	**0.9270**	**96.78**	**97.15**	**3179**	**0.8162**	**83.18**	**93.61**	**3755**	**0.8524**	**89.91**

where

$$\mathrm{PRE} = \frac{(\mathrm{TP}+\mathrm{FP})\cdot \mathrm{RC} + (\mathrm{TN}+\mathrm{FN})\cdot \mathrm{RU}}{(\mathrm{FP}+\mathrm{FN}+\mathrm{TP}+\mathrm{TN})^2} \tag{20}$$

where RC and RU represent the number of pixels in the reference ground truth that are actually changed and unchanged, respectively.

3) Overall Error: This metric measured the number of misclassified samples by the model, indicating the model's error rate across all samples

$$\mathrm{OE} = \mathrm{FP} + \mathrm{FN} \tag{21}$$

4) F1 Score: This comprehensive metric evaluates the model's performance, particularly in imbalanced datasets, by considering both Precision and Recall. It is the harmonic mean of these two metrics, providing a balance between accuracy and completeness. A higher $F1$ score indicates better classification performance. The calculation method is as follows:

$$F1 = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{22}$$

where

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{23}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{24}$$

C. Implementation Details

The experiments in this study were conducted using the PyTorch [52] framework on an NVIDIA GTX 3090 GPU. We used the SGD optimizer with a batch size of 32 and 500 training iterations. The optimizer's momentum was set to 0.5, and the initial learning rate was fixed at 0.001. To adapt to the model's varying sensitivity to the learning rate during different stages of training, we implemented a learning rate scheduler. This dynamic adjustment aimed to accelerate convergence and improve model performance.

For the training set, we randomly selected 50 samples from both the change and no-change classes in each dataset, ensuring balanced learning from both categories. Since adjacent pixels in HSIs typically correspond to the same ground object and contain similar information, we used 9 × 9 patches as input to extract differential features. In addition, we utilized the THOP library to calculate 1.84 M multiply–accumulate operations (MACs) and 6.55 G parameters for our model. The inference times for our model are 157.12 s on the SantaBarbara dataset, 60.88 s on the BayArea dataset, 17.81 s on the River dataset, and 8.90 s on the China dataset.

D. Experimental Result and Analysis

To highlight the effectiveness of our proposed method, we performed comparative experiments against eight representative CD algorithms. These algorithms are categorized as follows: traditional unsupervised methods that do not require labeled data (CVA [23], RCVA [53], C2VA [54], PACKeans [55], and TDRD [56]), traditional supervised methods (SVM), unsupervised deep learning methods (KPCAMNet [57], DCVA [58], and SCIU), and supervised deep learning methods that rely on labeled data and leverage deep learning networks for feature learning and CD (GETNET [49], TDSSC [59], DSFANet [30], S^2AN [41], MSCSCNet [60], and DCENet [61]).

1) Results on SantaBarbara Dataset: The visual and quantitative evaluation results on the SantaBarbara dataset, as shown in Fig. 8 and Table I, respectively, highlight several limitations of traditional unsupervised algorithms such as PACKeans, C2VA, and TDRD. These methods suffer from significant FNs, FPs, and irregular change boundaries, particularly in regions with blurred transitions or minimal spectral differences. As a result, their CD outputs are often spotty and discontinuous. Specifically, PACKeans struggles with spatial consistency in complex or noisy backgrounds, leading to fragmented change areas. Similarly, CVA and RCVA demonstrate moderate performance, with noise concentrated around change regions, causing discontinuities in detection. This is partly due to CVA's reliance on pixelwise spectral distances, which fails to leverage spatial context and makes it more susceptible to noise. While RCVA addresses some of these issues, it still exhibits high error rates. In contrast, traditional supervised

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

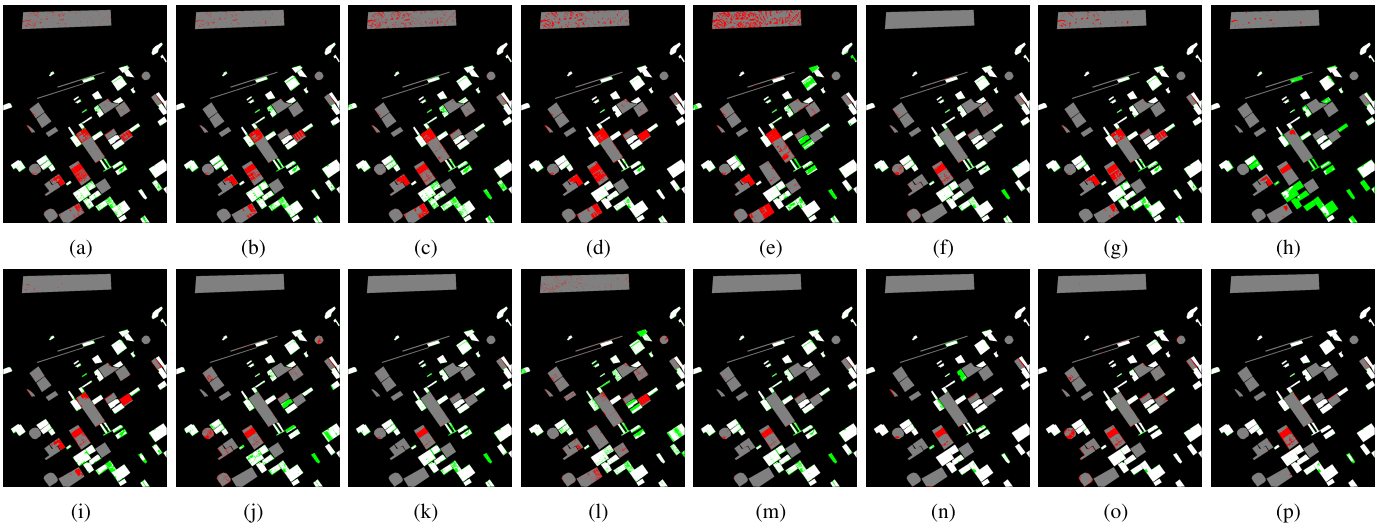


Fig. 8. Results on SantaBarbara dataset. The red indicates FP pixels and the green indicates missed detection pixels. (a) CVA. (b) RCVA. (c) C2VA. (d) TDRD. (e) PACKeans. (f) SVM. (g) KPCAMNet. (h) DCVA. (i) SCIU. (j) GETNET. (k) TDSSC. (l) DSFANet. (m) S^2AN. (n) MSCSCNet. (o) DCENet. (p) Ours.

methods like SVM and unsupervised deep learning approaches such as SCIU offer more coherent change boundaries, though some spotty noise remains. This indicates that incorporating supervised information or using deep learning to extract more complex features from HSIs can improve performance. Supervised deep learning models, including TDSSC, S^2AN, MSCSCNet, and DCENet, deliver comparable results, with stronger feature-capturing capabilities. However, they still face challenges in areas with subtle changes, likely due to limitations in network design or insufficient training data. Our proposed method, in contrast, produces CD maps with clear boundaries, significantly reducing noise and errors. It achieves higher spatial consistency and detection accuracy, demonstrating that our ACFHAN and semisupervised augmentation approach substantially outperform both traditional and deep learning methods on the SantaBarbara dataset.

2) Results on BayArea Dataset: The visual and quantitative comparative results on the BayArea dataset, as shown in Fig. 9 and Table I, respectively, highlight several key findings. The results from CVA, C2VA, and RCVA often suffer from significant FNs and FPs, particularly near complex building and land cover boundaries. While TDRD shows improvements in robustness, it does not fully address these noise issues, which remain prominent. PACKeans, which employs clustering for CD, reduces noise to some extent but still struggles with blurred and fragmented boundaries. The SVM method, based on supervised learning, performs well when trained on high-quality data but is slightly outperformed by deep learning methods. Among these, the deep learning method TDSSC stands out, achieving an $F1$ score of 94.39. It excels in detecting natural changes in suburban areas, where spatial consistency constraints help mitigate some of the noise. However, in high-density urban areas, local discontinuities and FNs remain. S^2AN, by incorporating deeper network architectures and modeling spatial consistency, significantly reduces FNs and FPs. However, its use of convolutional methods to compute attention scores results in blurring at the contour edges. Overall, the proposed method outperforms all other competitive algorithms, achieving the highest OA, KC, and $F1$ scores, along with the lowest OE score. Subjective evaluations further demonstrate that the proposed method effectively detects both significant and subtle changes.

3) Results on River Dataset: The visual and quantitative comparative results on the River dataset, shown in Fig. 10 and Table I, respectively, highlight significant FNs and FPs for CVA, C2VA, and TDRD. The minimal spectral differences around the river make it challenging for CVA to distinguish between change areas and background noise, leading to noticeable noise along the river's edges. While C2VA improves on CVA by incorporating spatial information, it still struggles to maintain spatial consistency in complex or noisy backgrounds, resulting in fragmented CD. In contrast, PACKeans performs better on the River dataset, showing a notable reduction in spots and an improvement in boundary continuity, achieving an OA of 95.56. However, when temporal changes are subtle, PACKeans may miss some changes, causing visible boundary irregularities and local noise in meandering river sections or dense vegetation areas. Deep learning methods such as KPCAMNet, S^2AN, DSFANet, and DCENet also perform well on the River dataset. KPCAMNet enhances CD by combining PCA with CNNs, improving both spectral feature reduction and spatial feature learning. DCVA strengthens spectral information extraction, while DSFANet utilizes a dual-branch convolutional network with multiscale features to further boost CD performance. Despite their advantages, these methods may still misclassify subtle changes due to the high-dimensional complexity of HSIs, leading to occasional small spots and misdetections. S^2AN utilizes self-attention to dynamically adjust feature weights, improving focus on key change areas while suppressing noise. However, while the attention method enhances feature selectivity, it may still struggle to capture local details, resulting in some misclassified regions in the detection images. Our proposed method, in contrast, effectively mitigates spectral redundancy and captures critical change information. The results on the River dataset demonstrate an OA of 97.15, with more precise boundaries for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

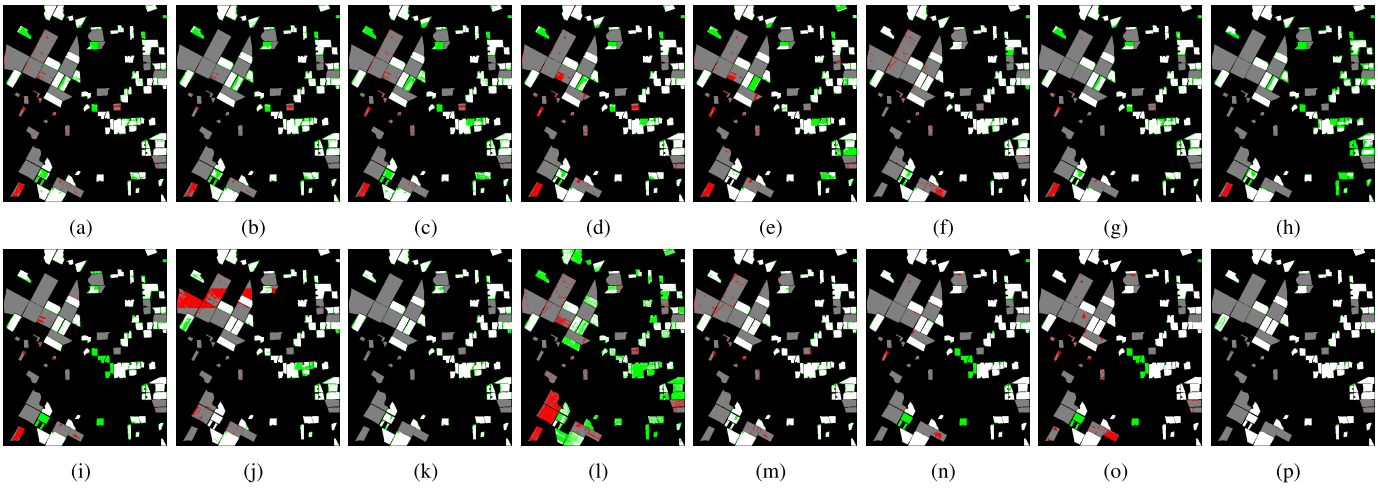


Fig. 9. Results on BayArea dataset. The red indicates FP pixels and the green indicates missed detection pixels. (a) CVA. (b) RCVA. (c) C2VA. (d) TDRD. (e) PACKeans. (f) SVM. (g) KPCAMNet. (h) DCVA. (i) SCIU. (j) GETNET. (k) TDSSC. (l) DSFANet. (m) S^2AN. (n) MSCSCNet. (o) DCENet. (p) Ours.

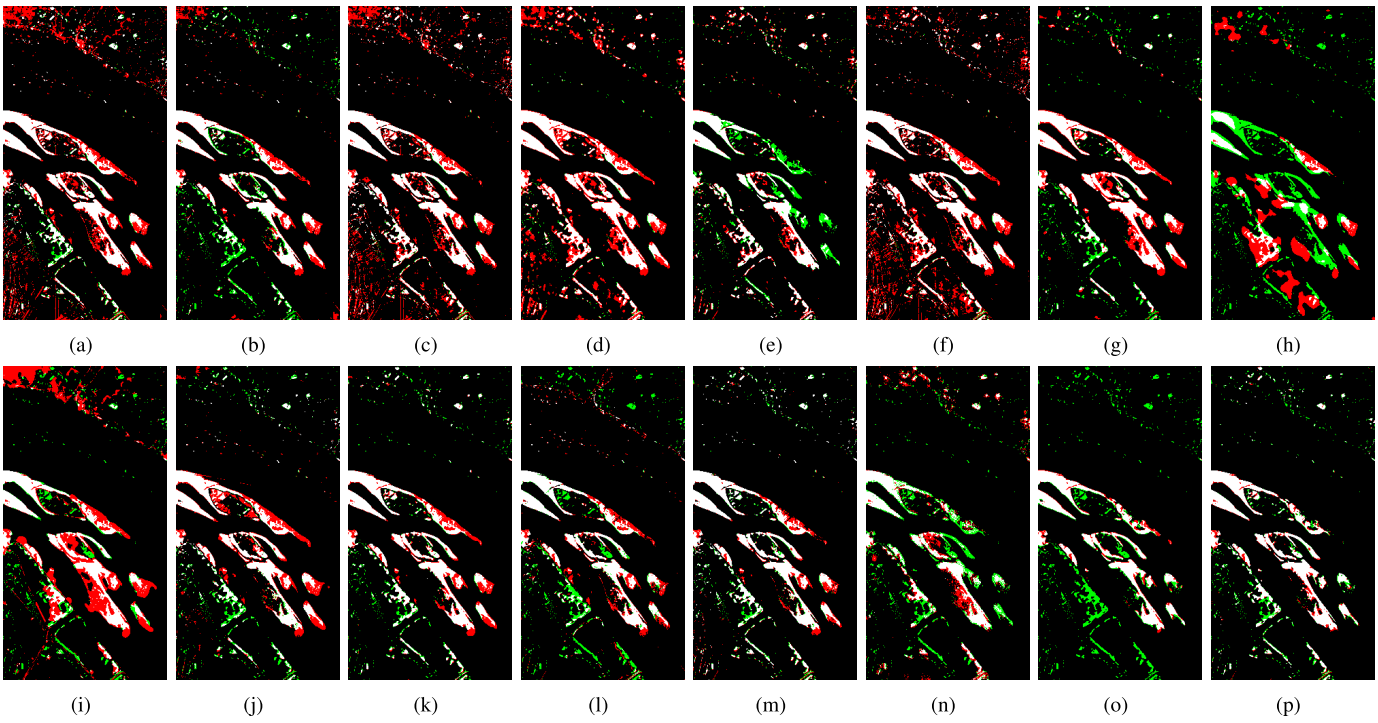


Fig. 10. Results on River dataset. The red indicates FP pixels and the green indicates missed detection pixels. (a) CVA. (b) RCVA. (c) C2VA. (d) TDRD. (e) PACKeans. (f) SVM. (g) KPCAMNet. (h) DCVA. (i) SCIU. (j) GETNET. (k) TDSSC. (l) DSFANet. (m) S^2AN. (n) MSCSCNet. (o) DCENet. (p) Ours.

change areas, significantly reduced FPs and FNs, and particularly strong performance in complex riverine environments.

4) Results on China Dataset: The visual and quantitative comparative results on the China dataset, shown in Fig. 11 and Table I, respectively, demonstrate that traditional unsupervised algorithms, including CVA, RCVA, C2VA, and PACKeans, produce a considerable amount of FPs (red regions) and FNs (green regions), particularly in areas with blurred boundaries. While TDRD shows slight improvements over these methods, achieving an approximately 1% increase in detection accuracy, it continues to suffer from notable FP and FN rates, especially in complex scenarios. Traditional supervised algorithms such as SVM achieve a reduction in FPs and FNs compared to unsupervised methods. However, they still exhibit significant FNs in regions with subtle changes or unclear boundaries, highlighting their limited adaptability to complex scenarios. Unsupervised deep learning methods, on the other hand, appear to better exploit spectral and spatial information. However, their lack of supervised guidance leads to extensive FNs in certain areas, reflecting insufficient robustness in handling complex change patterns. Supervised deep learning methods outperform the aforementioned approaches, achieving noticeable improvements in accuracy. Nevertheless, challenges remain in detecting fine-grained boundaries and subtle changes. For instance, results from GETNET, S^2AN, MSCSCNet, and DCENet still display significant FPs

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

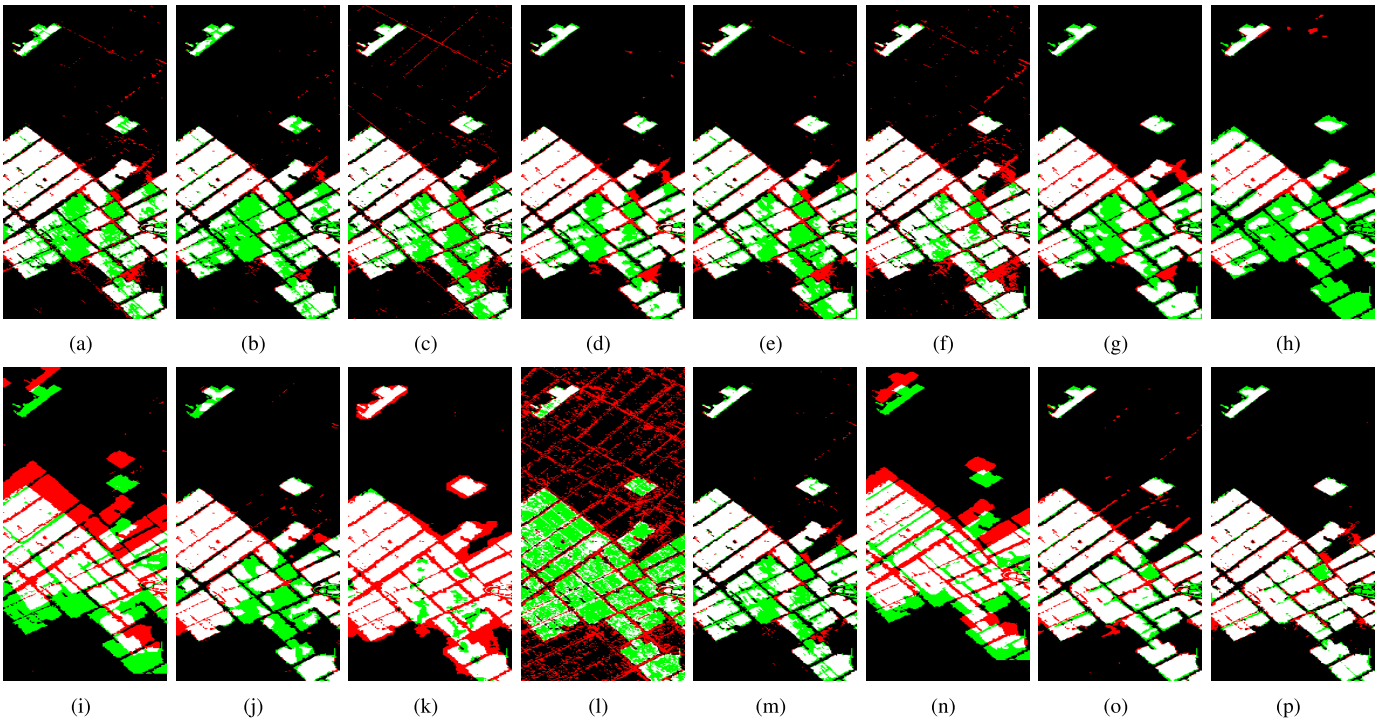


Fig. 11. Results on China dataset. The red indicates FP pixels and the green indicates missed detection pixels. (a) CVA. (b) RCVA. (c) C2VA. (d) TDRD. (e) PACKeans. (f) SVM. (g) KPCAMNet. (h) DCVA. (i) SCIU. (j) GETNET. (k) TDSSC. (l) DSFANet. (m) S^2AN. (n) MSCSCNet. (o) DCENet. (p) Ours.

and FNs. Notably, MSCSCNet exhibits a noticeable overall shift in detection results, further impacting its performance. In contrast, our proposed method demonstrates superior performance, accurately detecting changes even in regions with blurred boundaries and subtle variations. It significantly reduces FPs and FNs, delivers clear and consistent boundaries, and achieves strong spatial coherence. These results underscore the robustness and adaptability of our approach, particularly in addressing complex scenarios and detecting subtle changes, thereby validating its effectiveness and practical applicability.

E. Ablation Study and Sensitivity Analysis

To validate the effectiveness of the proposed method, we conduct ablation studies across all datasets in this section. And due to the varying scales of changes in the River dataset, which can encompass different scales of ground objects, we choose to conduct sensitivity analysis on patch size for this dataset. We first demonstrate the indispensable role of each module in our proposed network, then analyze the advantages of our superpixel-based semisupervised label augmentation method, and finally validate that our proposed CFA module can reduce the model's sensitivity to patch size to some extent.

1) Attention Module: In our proposed model, the key components are the S2HSA module, designed for extracting spatial and channel features, and the CFA module, which enhances the central point features. To evaluate the role of each module, we conducted ablation experiments on all datasets. The results are shown in Table II. The table presents the performance of four different methods in the task of HSIs CD, including the following: using no attention modules, using only the S2HSA module, using only the CFA module, and the proposed method that combines both modules. Notably, we set the patch size to 9 uniformly and trained the models using label samples enhanced by our superpixel-based semisupervised label augmentation method.

TABLE II
QUANTITATIVE EVALUATION OF EXPERIMENTAL RESULTS OBTAINED BY METHODS WITH DIFFERENT ATTENTION ON FOUR DATASETS

Dataset	CNN	S2HSA	CFA	OA	OE	KC	F1
SantaBarbara	✓	-	-	95.54	5916	0.9092	94.44
	-	✓	-	96.05	5231	0.9178	95.06
	-	-	✓	95.86	5490	0.9131	94.72
	✓	✓	✓	**96.38**	**4801**	**0.9245**	**95.46**
BayArea	✓	-	-	95.32	3070	0.9054	95.79
	-	✓	-	96.09	2569	0.9207	96.50
	-	-	✓	95.95	2657	0.9180	96.37
	✓	✓	✓	**96.40**	**2364**	**0.9270**	**96.78**
River	✓	-	-	96.04	4418	0.7204	74.15
	-	✓	-	96.55	3852	0.7623	78.08
	-	-	✓	96.44	3973	0.7703	78.97
	✓	✓	✓	**97.15**	**3179**	**0.8162**	**83.18**
China	✓	-	-	91.71	4877	0.8171	87.96
	-	✓	-	92.52	4396	0.8263	88.08
	-	-	✓	93.12	4046	0.8439	89.49
	✓	✓	✓	**93.61**	**3755**	**0.8524**	**89.91**

As can be observed, the introduction of the S2HSA module significantly improves the overall performance of the model. This indicates that the S2HSA module effectively enhances feature selection and extraction, reducing misclassification and achieving better performance than the simple CNN architecture. The inclusion of the CFA module further improves

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUANTITATIVE EVALUATION OF EXPERIMENTAL RESULTS OBTAINED BY DIFFERENT DATA AUGMENTATION METHODS ON FOUR DATASETS

Model	SantaBarbara				BayArea				River				China			
	OA	OE	KC	F1	OA	OE	KC	F1	OA	OE	KC	F1	OA	OE	KC	F1
only labeled	95.69	5711	0.9094	94.48	94.75	3956	0.8764	94.80	95.75	4747	0.7298	75.31	91.98	4714	0.8141	87.26
USSM	93.90	8083	0.8729	92.36	93.87	4025	0.8760	94.48	92.90	7926	0.6380	67.59	83.07	9957	0.5490	62.44
SSSM	94.21	7672	0.8790	92.69	95.70	2823	0.9119	96.28	95.26	5285	0.7385	76.43	92.79	4241	0.8363	88.97
S^2AN	94.68	7047	0.8879	93.12	93.60	4203	0.8714	94.13	96.96	3395	0.8139	83.06	92.57	4368	0.8257	87.94
TSPLR	95.56	5879	0.9069	94.34	89.57	6846	0.7929	90.07	93.89	6821	0.6416	67.51	89.22	6339	0.7506	82.94
CBW-MSSANet	92.11	10453	0.8349	90.00	90.07	6519	0.8024	90.60	95.14	5425	0.7252	75.18	89.92	5929	0.7623	83.48
MCS^{4}CD	95.34	6171	0.9027	94.12	93.28	4411	0.8648	93.86	94.54	6091	0.7011	73.07	92.78	4246	0.8337	88.66
Ours	**96.38**	**4801**	**0.9245**	**95.46**	**96.40**	**2364**	**0.9270**	**96.78**	**97.15**	**3179**	**0.8162**	**83.18**	**93.61**	**3755**	**0.8524**	**89.91**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 12. Sensitivity analysis about the patch size on the River Dataset. (a)–(e) CNN (patch size = 13, 15, 27, 31, 39). (f)–(j) ACFHAN (patch size = 13, 15, 27, 31, 39).

the performance metrics compared to the S2HSA module alone, particularly in terms of the KC and $F1$ score. This demonstrates that the CFA module is highly effective in capturing crucial features from HSIs data and in distinguishing between change and non-change areas. Ultimately, the proposed method, which integrates both the S2HSA and CFA modules, achieves the best results across all evaluation metrics, validating the effectiveness and superiority of this method in HSIs CD.

2) Label Augmentation Method: Combining supervised and unsupervised learning, semisupervised learning utilizes a small amount of labeled data alongside a large volume of unlabeled data for training, enhancing the model's predictive capabilities. Here, we compared the performance of using only labeled data without data augmentation with that of our proposed unsupervised sample selection method (USSM), the supervised sample selection method (SSSM), and the method that combines both approaches. The results using different methods are shown in Table III, where we can see a significant improvement in performance with the method that combines both approaches compared to other supervised and unsupervised methods, further validating the advantages of semisupervised learning. Our final proposed method, which integrates all approaches, achieves the best results by effectively extracting information from both HSIs and their difference maps, demonstrating the effectiveness of our proposed strategy.

Additionally, we compared our proposed data augmentation strategy with the sample augmentation techniques introduced in four recent algorithms. Specifically, to be fair, the augmented training samples of these algorithms will be trained uniformly on our proposed ACFHAN. We incorporated two unsupervised data augmentation methods, TSPLR [62] and CBW-MSSANet [63], and two supervised data augmentation methods, S^2AN and MCS^{4}CD [64], to further assess the robustness and effectiveness of our approach. The results are presented in Table III. From the table, it is clear that, compared to the fuzzy C-means clustering method in the TSPLR algorithm, our method improves accuracy by 3.26% on the River dataset. This improvement suggests that our approach is more effective in handling edge and fine-detail labeling, particularly in regions where land object boundaries experience complex or subtle changes. CBW-MSSANet, which selects high-confidence training samples based on K-means clustering and centroid distance, struggles with misclassifications in areas with more intricate land object changes, leading to reduced performance. In contrast, S^2AN and MCS^{4}CD introduce a limited amount of labeled data for data augmentation, effectively mitigating the impact of incorrect labels. However, since their augmentation strategies are applied on a per-pixel basis and do not consider neighboring pixel information, their effectiveness is somewhat constrained. Our proposed method, by efficiently extracting information from both the HSIs and its difference maps, achieves the best performance, further demonstrating the superiority and effectiveness of our approach.

3) Patch Size Sensitivity Analysis: The experimental results of patch-based methods are often influenced by patch size. In our experiments, we found that the CFA module, by enhancing the feature weights of the central points, can to some extent reduce the dependence of patch-based methods on patch size. We evaluated the performance of a standard CNN model and our proposed model on the River dataset using patch sizes of 13, 15, 23, 31, and 39. To minimize the impact of inaccurate training samples, we selected 50 pixels each from the changed and unchanged pixel categories as training samples. The results are shown in Fig. 12. As patch size increases, it becomes evident that the CNN-generated images

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

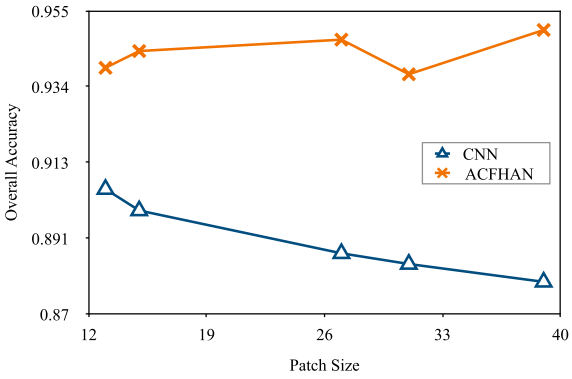


Fig. 13. OA curves obtained by training the basic CNN and our proposed ACFHAN model using different patch sizes, showing how OA varies with patch size.

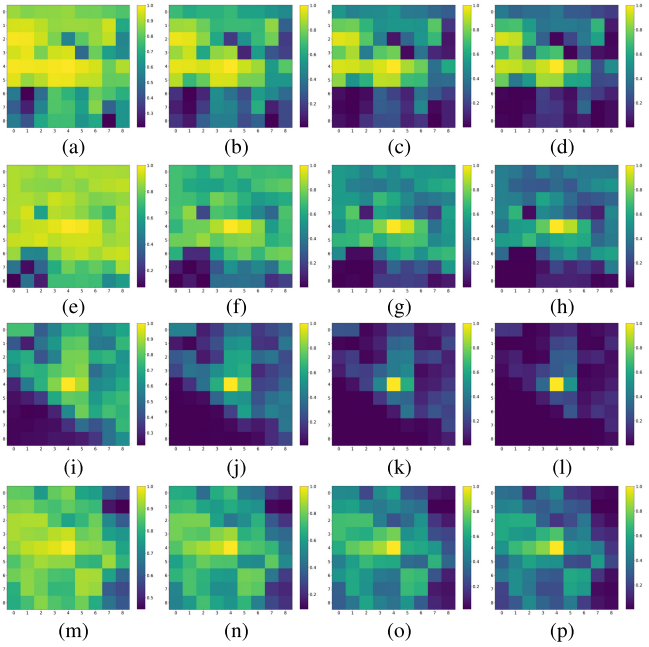


Fig. 14. Attention scores of the CFA module are visualized in heatmaps, varying across different layers. Panels (a)–(d), (e)–(h), (i)–(l), and (m)–(p) present the CFA attention score heatmaps for different patches after passing through layers 1 to 4 of the CFA module.

grow progressively coarser. In contrast, ACFHAN consistently preserves clear and distinct object boundaries at all patch sizes. This is visually illustrated in Fig. 13, which depicts the OA curves for both models across different patch sizes. ACFHAN maintains a stable OA around 0.94, while CNN's OA declines steadily from 0.91. Furthermore, ACFHAN consistently outperforms CNN, highlighting the effectiveness of the proposed attention-based method in enhancing model stability.

In addition, to further improve the interpretability of the proposed method, we randomly selected four 9×9 patches from the River dataset and visualized the attention score heatmaps of the CFA module, thereby analyzing its adaptive focus on the central region of these patches. The specific visual results are shown in Fig. 14. Panels Fig. 14(a)–(d), (e)–(h), (i)–(l), and (m)–(p) present the attention score heatmaps for each of the four patches after passing through 1 to 4 layers of the CFA module. These visualizations reveal that, as the number of layers increases, the attention weight of the central pixel progressively intensifies, while the attention weights of the surrounding pixels decrease adaptively, with the focus gradually concentrating on the central region. These findings further validate the effectiveness of the proposed adaptive CFA module.

V. Conclusion

In this article, we proposed a novel feature-enhanced ACFHAN designed to improve CD in HSIs. ACFHAN effectively enhances strongly correlated features relevant to CD while suppressing weakly correlated and irrelevant information. The network is composed of two key modules: S2HSA and CFA. The S2HSA module calculates both channel and SSA scores for the input feature maps, whereas the CFA module assesses the relationships between surrounding pixels and the central pixel. Notably, our results indicate that CFA reduces the model's sensitivity to patch size, thereby mitigating the potential negative effects of varying patch sizes during training. Furthermore, we introduced a superpixel-based data sample enhancement method by integrating superpixel segmentation with simple classification algorithms, significantly reducing the network's dependency on labeled training samples.

Although our proposed method provides a powerful solution for HSIs CD, there are still some inherent limitations and potential scenarios where the model's performance may be compromised. In future work, we will further investigate the issues our method currently faces, including label and data dependency, as well as the lack of flexibility and adaptability.

References

[1] B. Zhang et al., "Progress and challenges in intelligent remote sensing satellite systems," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1814–1822, 2022.

[2] S. Yuan et al., "Marine environmental monitoring with unmanned vehicle platforms: Present applications and future prospects," *Sci. Total Environ.*, vol. 858, Feb. 2023, Art. no. 159741.

[3] M. Kucharczyk and C. H. Hugenholtz, "Remote sensing of natural hazard-related disasters with small drones: Global trends, biases, and research opportunities," *Remote Sens. Environ.*, vol. 264, Oct. 2021, Art. no. 112577.

[4] T. Wellmann et al., "Remote sensing in urban planning: Contributions towards ecologically sound policies?" *Landscape Urban Planning*, vol. 204, Dec. 2020, Art. no. 103921.

[5] S. Luo, J. Yu, Y. Xi, and X. Liao, "Aircraft target detection in remote sensing images based on improved YOLOv5," *IEEE Access*, vol. 10, pp. 5184–5192, 2022.

[6] K. Perumal and R. Bhaskaran, "Supervised classification performance of multispectral images," 2010, *arXiv:1002.4046*.

[7] W. Sun, K. Ren, X. Meng, G. Yang, J. Peng, and J. Li, "Unsupervised 3-D tensor subspace decomposition network for spatial–temporal–spectral fusion of hyperspectral and multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5528917.

[8] Y. K. Chan and V. Koo, "An introduction to synthetic aperture radar (SAR)," *Progr. Electromagn. Res. B*, vol. 2, pp. 27–60, 2008.

[9] E. Hamidi, B. G. Peter, D. F. Mu noz, H. Moftakhari, and H. Moradkhani, "Fast flood extent monitoring with SAR change detection using Google Earth engine," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Jan. 2023, Art. no. 4201419.

[10] X. Wu, P. Gamba, J. Feng, R. Shang, X. Zhang, and L. Jiao, "A multitask framework for hyperspectral change detection and band reweighting with unbalanced contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5530213.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

[11] Z. Lv, M. Zhang, W. Sun, J. A. Benediktsson, T. Lei, and N. Falco, "Spatial-contextual information utilization framework for land cover change detection with hyperspectral remote sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4411911.

[12] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.

[13] G. Lu and B. Fei, "Medical hyperspectral imaging: A review," *J. Biomed. Opt.*, vol. 19, no. 1, Jan. 2014, Art. no. 010901.

[14] A. M. Martinez and A. C. Kak, "PCA versus LDA," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 23, no. 2, pp. 228–233, Feb. 2001.

[15] A. Song, J. Choi, Y. Han, and Y. Kim, "Change detection in hyperspectral images using recurrent 3D fully convolutional networks," *Remote Sens.*, vol. 10, no. 11, p. 1827, Nov. 2018.

[16] F. Huang, Y. Yu, and T. Feng, "Hyperspectral remote sensing image change detection based on tensor and deep learning," *J. Vis. Commun. Image Represent.*, vol. 58, pp. 233–244, Jan. 2019.

[17] M. Hu, C. Wu, B. Du, and L. Zhang, "Binary change guided hyperspectral multiclass change detection," *IEEE Trans. Image Process.*, vol. 32, pp. 791–806, 2023.

[18] X. Ou, L. Liu, B. Tu, G. Zhang, and Z. Xu, "A CNN framework with slow-fast band selection and feature fusion grouping for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524716.

[19] S. L. Al-Khafaji, J. Zhou, X. Bai, Y. Qian, and A. W. Liew, "Spectral–spatial boundary detection in hyperspectral images," *IEEE Trans. Image Process.*, vol. 31, pp. 499–512, 2022.

[20] R. Guan, Z. Li, X. Li, and C. Tang, "Pixel-superpixel contrastive learning and pseudo-label correction for hyperspectral image clustering," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 6795–6799.

[21] B. Manifold, S. Men, R. Hu, and D. Fu, "A versatile deep learning architecture for classification and label-free prediction of hyperspectral images," *Nature Mach. Intell.*, vol. 3, no. 4, pp. 306–315, Mar. 2021.

[22] X. Ren and J. Malik, "Learning a classification model for segmentation," in *Proc. 9th IEEE Int. Conf. Comput. Vis.*, vol. 1, Oct. 2003, pp. 10–17.

[23] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2006.

[24] S. Liu, L. Bruzzone, F. Bovolo, and P. Du, "A novel sequential spectral change vector analysis for representing and detecting multiple changes in hyperspectral images," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 4656–4659.

[25] V. Ortiz-Rivera, M. Vélez-Reyes, and B. Roysam, "Change detection in hyperspectral imagery using temporal principal components," *Proc. SPIE*, vol. 6233, pp. 368–377, May 2006.

[26] C. Wu, B. Du, and L. Zhang, "A subspace-based change detection method for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 815–830, Apr. 2013.

[27] F. Bovolo, L. Bruzzone, and M. Marconcini, "A novel approach to unsupervised change detection based on a semisupervised SVM and a similarity measure," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2070–2082, Jul. 2008.

[28] L. Alzubaidi et al., "Review of deep learning: Concepts, CNN architectures, challenges, applications, future directions," *J. Big Data*, vol. 8, no. 1, pp. 1–74, Mar. 2021.

[29] W. Dong, Y. Yang, J. Qu, S. Xiao, and Y. Li, "Local information-enhanced graph-transformer for hyperspectral image change detection with limited training samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509814.

[30] B. Du, L. Ru, C. Wu, and L. Zhang, "Unsupervised deep slow feature analysis for change detection in multi-temporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9976–9992, Dec. 2019.

[31] B. Yang, Y. Mao, L. Liu, X. Liu, Y. Ma, and J. Li, "From trained to untrained: A novel change detection framework using randomly initialized models with spatial–channel augmentation for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4402214.

[32] T. Zhan et al., "SSCNN-S: A spectral–spatial convolution neural network with Siamese architecture for change detection," *Remote Sens.*, vol. 13, no. 5, p. 895, Feb. 2021.

[33] Y. Wang et al., "Spectral–spatial–temporal transformers for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536814.

[34] J. M. Haut, M. E. Paoletti, J. Plaza, A. Plaza, and J. Li, "Visual attention-driven hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8065–8080, Oct. 2019.

[35] L. Mou and X. X. Zhu, "Learning to pay attention on spectral domain: A spectral attention module-based convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 110–122, Jan. 2020.

[36] B. Fang, Y. Li, H. Zhang, and J. Chan, "Hyperspectral images classification based on dense convolutional networks with spectral-wise attention mechanism," *Remote Sens.*, vol. 11, no. 2, p. 159, Jan. 2019.

[37] L. Wang, L. Wang, Q. Wang, and P. M. Atkinson, "SSA-SiamNet: Spectral–spatial-wise attention-based Siamese network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5510018.

[38] Q. Li et al., "A superpixel-by-superpixel clustering framework for hyperspectral change detection," *Remote Sens.*, vol. 14, no. 12, p. 2838, Jun. 2022.

[39] W. Dong et al., "Abundance matrix correlation analysis network based on hierarchical multihead self-cross-hybrid attention for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501513.

[40] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.

[41] M. Gong et al., "A spectral and spatial attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521614.

[42] Y. Li et al., "CBANet: An end-to-end cross-band 2-D attention network for hyperspectral change detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513011.

[43] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[44] L. Cai, X. Xu, J. H. Liew, and C. Sheng Foo, "Revisiting superpixels for active learning in semantic segmentation with realistic annotation costs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10983–10992.

[45] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[46] N. Otsu et al., "A threshold selection method from gray-level histograms," *Automatica*, vol. 11, nos. 285–296, pp. 23–27, 1975.

[47] K. Huang, S. Li, X. Kang, and L. Fang, "Spectral–spatial hyperspectral image classification based on KNN," *Sens. Imag.*, vol. 17, no. 1, pp. 1–13, Dec. 2016.

[48] J. López-Fandiño, D. B. Heras, F. Argüello, and M. D. Mura, "GPU framework for change detection in multitemporal hyperspectral images," in *Proc. 10th Int. Symp. High-Level Parallel Program. Appl.*, vol. 47, Dec. 2017, pp. 272–292.

[49] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2018.

[50] M. Hasanlou and S. T. Seydi, "Hyperspectral change detection: An experimental comparative study," *Int. J. Remote Sens.*, vol. 39, no. 20, pp. 7029–7083, Oct. 2018.

[51] G. H. Rosenfield and K. Fitzpatrick-Lins, "A coefficient of agreement as a measure of thematic classification accuracy," *Photogram. Eng. Remote Sens.*, vol. 52, no. 2, pp. 223–227, 1986.

[52] A. Paszke et al., "Automatic differentiation in PyTorch," in *Proc. NIPS-W*, 2017.

[53] F. Thonfeld, H. Feilhauer, M. Braun, and G. Menz, "Robust change vector analysis (RCVA) for multi-sensor very high resolution optical satellite data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 50, pp. 131–140, Aug. 2016.

[54] F. Bovolo, S. Marchesi, and L. Bruzzone, "A framework for automatic and unsupervised detection of multiple changes in multitemporal images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 6, pp. 2196–2212, Jun. 2012.

[55] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and k-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

[56] Z. Hou, W. Li, R. Tao, and Q. Du, "Three-order Tucker decomposition and reconstruction detector for unsupervised hyperspectral change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6194–6205, 2021.

[57] C. Wu, H. Chen, B. Du, and L. Zhang, "Unsupervised change detection in multitemporal VHR images based on deep kernel PCA convolutional mapping network," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 12084–12098, Nov. 2022.

[58] S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised deep change vector analysis for multiple-change detection in VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3677–3693, Jun. 2019.

[59] T. Zhan et al., "TDSSC: A three-directions spectral–spatial convolution neural network for hyperspectral image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 377–388, 2020.

[60] J. Qu, P. Yang, W. Dong, X. Zhang, and Y. Li, "A semi-supervised multiscale convolutional sparse coding-guided deep interpretable network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5531314.

[61] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and X. Gao, "DCENet: Diff-feature contrast enhancement network for semi-supervised hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511514.

[62] S. Fang, C. Qi, S. Yang, Z. Li, W. Wang, and Y. Wang, "Unsupervised SAR change detection using two-stage pseudo labels refining framework," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[63] X. Ou, L. Liu, B. Tu, L. Qing, G. Zhang, and Z. Liang, "CBW-MSSANet: A CNN framework with compact band weighting and multiscale spatial attention for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507618.

[64] L. Liu, D. Hong, L. Ni, and L. Gao, "Multilayer cascade screening strategy for semi-supervised change detection in hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1926–1940, 2022.



Fenlong Jiang (Member, IEEE) received the B.Eng. and Ph.D. degrees in electronic science and technology from Xidian University, Xi'an, China, in 2017 and 2022, respectively.

Since 2023, he has been a Lecturer with the School of Computer Science and Technology, Xidian University. His research interests include deep learning, computer vision, and remote sensing image processing.



Shining Zhang received the B.Eng. degree from Xidian University, Xi'an, China, in 2021, where she is currently pursuing the M.Eng. degree in electronic science and technology.

Her research interests include deep learning and remote sensing change detection.



Mingyang Zhang (Member, IEEE) received the B.Eng. degree in automation and the Ph.D. degree in pattern recognition and intelligent system from Xidian University, Xi'an, China, in 2012 and 2018, respectively.

Since 2018, he has been a Lecturer with the School of Electronic Engineering, Xidian University, and was promoted to an Associate Professor in 2022. His research interests include computational intelligence and remote sensing image understanding.



Maoguo Gong (Fellow, IEEE) received the B.Eng. and Ph.D. degrees from Xidian University, Xi'an, China, in 2003 and 2009, respectively.

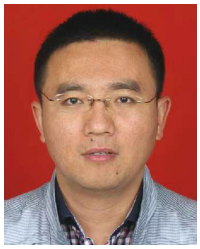
Since 2006, he has been a Teacher with Xidian University, where he was promoted to an Associate Professor and a Full Professor in 2008 and 2010, respectively, with exceptive admission. He has authored or co-authored over 100 articles in journals and conferences. He holds over 20 granted patents as the first inventor. He is leading or has completed over 20 projects as the Principal Investigator, funded by the National Natural Science Foundation of China and the National Key Research and Development Program of China. His research interests are broadly in the area of computational intelligence, with applications to optimization, learning, data mining, and image understanding.

Dr. Gong is the Executive Committee Member of the Chinese Association for Artificial Intelligence and a Senior Member of the Chinese Computer Federation. He was a recipient of the prestigious National Program for Support of the Leading Innovative Talents from the Central Organization Department of China, the Leading Innovative Talent in Science and Technology from the Ministry of Science and Technology of China, the Excellent Young Scientist Foundation from the National Natural Science Foundation of China, the New Century Excellent Talent from the Ministry of Education of China, and the National Natural Science Award of China. He is an Associate Editor or an Editorial Board Member for over five journals, including IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION and IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS.



Yu Zhou received the B.Eng. and M.S. degrees from Air Force Engineering University, Xi'an, China, in 2005 and 2008, respectively, and the Ph.D. degree from the National University of Defense Technology, Changsha, China, in 2013.

Since 2021, he has been a Teacher with Xidian University, where he is currently a Professor. His research interests include computational intelligence and multiparty collaborative learning.



Wei Zhao (Member, IEEE) received the B.S., M.S., and Ph.D. degrees from Xidian University, Xi'an, China, in 2002, 2005 and 2015, respectively.

He is currently a Professor with the School of Computer Science and Technology, Xidian University. His research direction is pattern recognition and intelligent systems, with specific interests in attributed graph mining and search, machine learning, signal processing, and precision guiding technology.



Ziyu Guan (Member, IEEE) received the B.S. and Ph.D. degrees in computer science from Zhejiang University, Hangzhou, China, in 2004 and 2010, respectively.

He had worked as a Research Scientist with the University of California at Santa Barbara, CA, USA, from 2010 to 2012, and as a Professor with the School of Information and Technology, Northwest University, Xi'an, China, from 2012 to 2018. He is currently a Professor with the School of Computer Science and Technology, Xidian University, Xi'an. His research interests include attributed graph mining and search, machine learning, expertise modeling and retrieval, and recommender systems.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.